



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name :Quantum Quotients

Team Leader Name :Golla Bhargava Teja

Problem Statement :AI-Driven Automated Crop type, Moisture Stress Detection and irrigation advisory Across Growth Stages Using Moderate Resolution Spectral Signatures (Optical & Microwave Satellite Data)

Team Members

Team Leader: Golla Bhargava Teja
Member -1

Name:Golla Bhargava Teja
College:CMR College of Engineering and
Technology

Team Member-2:

Name:Shravya Dharji
College:CMR College of Engineering and
Technology

Team Member-3:

Name:Maduapthi saipavan
College:CMR College of Engineering and
Technology

Project Jalachara— AI-Powered Predictive Water Governance for Canal Command Areas

❑ How Will It Solve the Problem?

- Forecast field-level water deficits using satellite data and AI.
- Convert water deficit into expected yield loss for better prioritization.
- Allocate limited canal water fairly using the Water Equity Index.
- Generate actionable irrigation advisories before crop damage occurs.
- Provide a dashboard for planners, canal authorities, and farmers.

From observation → prediction → action.

➤ USP

India's First Predictive Water Governance System

- ❖ Yield Loss Prevention Score
- ❖ Water Equity Index
- ❖ 8-Day Early Warning System

Built using ISRO satellite data (EOS-04, LISS series) and designed to scale with NISAR.

Existing Systems	Jalachara
Detect stress after visible crop damage	Predict stress 8 days before visible symptoms
Optimize only for yield	Optimize for yield + equity + early warning
Ignore tail-end farmer deprivation	Water Equity Index protects vulnerable farmers
Report water deficit in mm	Reports expected Yield Loss (%)
Monitoring tool	Decision-support and water governance system

Most solutions monitor. Jalachara governs.

"Jalachara doesn't just identify stressed crops—it predicts future stress, estimates yield impact, and guides equitable water allocation before losses occur."

Jalachara: Core Capabilities

SATELLITE TECHNOLOGY FOR SMART AGRICULTURE

From Space to Soil: Enabling Data-Driven Farming for a Sustainable Future

IN 3 SIMPLE STEPS



SATELLITE DATA ACQUISITION
High-resolution imagery from space



DATA PROCESSING & ANALYSIS
Advanced algorithms extract insights



ACTIONABLE ADVISORIES
Timely information for smarter farm decisions



Crop Monitoring
Real-time assessment of crop health and growth



Water Management
Optimized irrigation and resources



Risk Forecasting
Early warning for pests, diseases and weather



Data-Driven Decisions
Actionable insights for better yield and income



RICE



SEEDLING | TILLERING | FLOWERING | MATURITY

Better yield | Efficient water use

COTTON



SEEDLING | VEGETATIVE | FLOWERING | MATURITY

Healthy crop | Higher productivity

MAIZE



SEEDLING | VEGETATIVE | FLOWERING | MATURITY

Timely advisory | Maximum yield

WHEAT



SEEDLING | TILLERING | FLOWERING | MATURITY

Quality grain | Increased income

PULSES



SEEDLING | VEGETATIVE | FLOWERING | MATURITY

Protein rich | Soil enrichment



WIDE AREA COVERAGE
Monitor millions of hectares



ALL WEATHER CAPABILITY
Day or night, rain or shine



HIGH SPATIAL RESOLUTION
Field-level insights



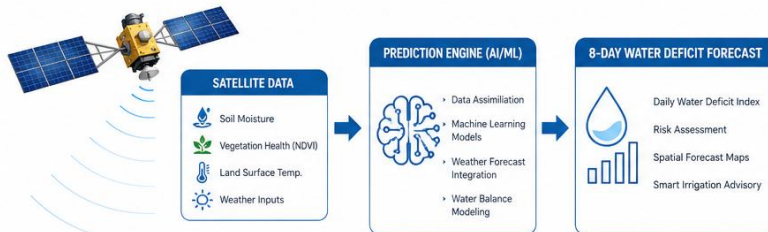
SUSTAINABLE AGRICULTURE
For food security and a better tomorrow



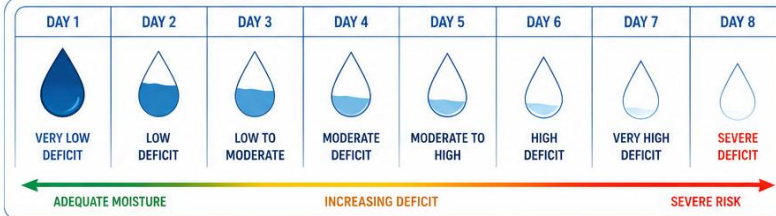
SATELLITE-POWERED WATER DEFICIT FORECAST

8-DAY OUTLOOK FOR SMART IRRIGATION

From Satellite Data to Actionable Insights



8-DAY WATER DEFICIT FORECAST (FIELD LEVEL)



SMART IRRIGATION ADVISORY

No irrigation required in next 3 days.
Monitor closely from Day 4.
Irrigation recommended on Day 6–7
for red and orange zones.



OPTIMIZED
WATER USE



HEALTHIER
CROPS



HIGHER
PROFITABILITY



SUSTAINABLE
AGRICULTURE

KEY BENEFITS



BETTER DECISIONS

Plan irrigation
with confidence



SAVE WATER

Irrigate only
when needed



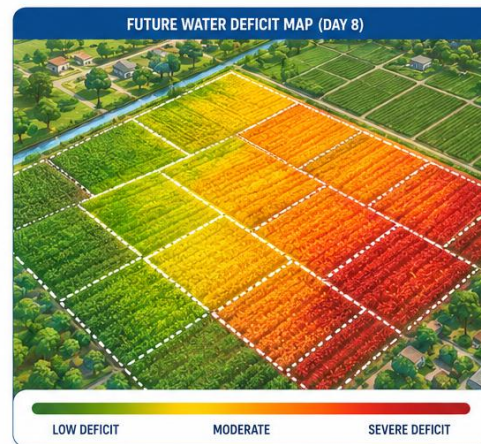
LOWER COSTS

Reduce input
and energy costs



HIGHER YIELD

Healthy crops,
higher productivity



Satellite
Imagery



Weather
Forecast



IoT Sensors
(Optional)



Field
Data

DATA SOURCES



AI-DRIVEN. DATA-POWERED. FUTURE-READY FARMING. | Plan Ahead. Save Water. Grow Better.

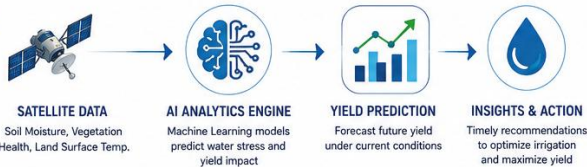


AGRICULTURAL AI

SMART INSIGHTS. HIGHER YIELDS.

PREDICT • PREVENT • PROTECT

Turning Water Intelligence into Higher Productivity



EXPECTED YIELD LOSS

14%

Due to projected water deficit in the next 8 days

HEALTHY CROP – OPTIMAL WATER

EXPECTED YIELD

6.2 TONS/HA

WATER STATUS

ADEQUATE

CROP CONDITION

HEALTHY

RICE

WHEAT

MAIZE

COTTON

SOYBEAN

VS

DROUGHT-AFFECTED CROP – WATER DEFICIT

EXPECTED YIELD

5.3 TONS/HA

WATER STATUS

DEFICIT

CROP CONDITION

STRESSED

RICE

WHEAT

MAIZE

COTTON

SOYBEAN

WATER DEFICIT TO YIELD IMPACT



Expected Yield Loss = **0.9 TONS/HA** in the next 8 days

14%

IRRIGATION RECOMMENDATION – ACT NOW TO PREVENT YIELD LOSS



Timely irrigation can recover up to **86%** of potential yield loss.

KEY BENEFITS



SAVE WATER
Irrigate only when needed



REDUCE COSTS
Lower input and energy costs



PROTECT YIELDS
Prevent stress, maximize productivity



DATA-DRIVEN
AI-powered insights for better decisions

DATA SOURCES



Satellite Imagery (NDVI, LST, Soil Moisture)



Weather Data (Forecast & Historical)



IoT Sensors (Soil & Weather)



Field Records (Crop & Yield Data)



Smart Decisions Today.
Stronger Harvests Tomorrow.



EQUITABLE WATER DISTRIBUTION FOR SUSTAINABLE AGRICULTURE

AI-POWERED CANAL IRRIGATION MANAGEMENT

Fair Water. Stronger Farms. Sustainable Future.



SMART FARMING EARLY WARNING SYSTEM

AI PREDICTS. FARMER ACTS. CROPS PROTECTED.



AI-POWERED AGRICULTURAL INTELLIGENCE
Turning Data Into Decisions
Securing Today's Harvest, Sustaining Tomorrow

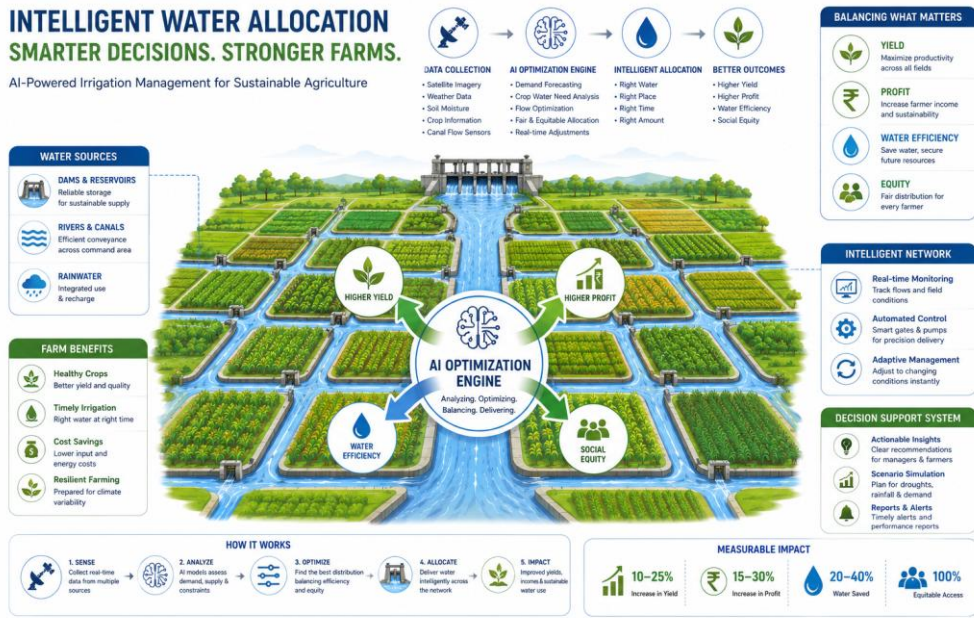


EARLY WARNING TODAY. BETTER HARVEST TOMORROW. | SMART DECISIONS. | STRONGER FARMS. | SUSTAINABLE FUTURE.

INTELLIGENT WATER ALLOCATION

SMARTER DECISIONS. STRONGER FARMS.

AI-Powered Irrigation Management for Sustainable Agriculture



DATA-DRIVEN. AI-POWERED. EQUITABLE. SUSTAINABLE.

Every Drop Counts. Every Farmer Matters.

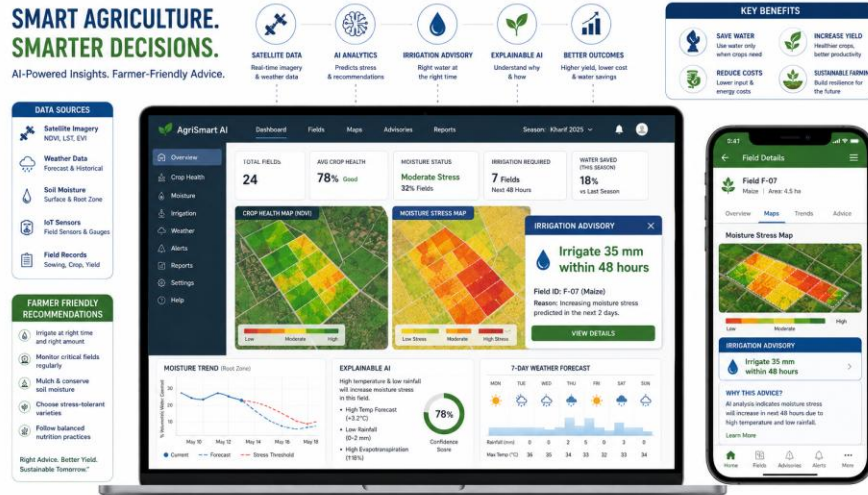
Together for a Water-Secure Future.

SMART WATER MANAGEMENT FOR SMART AGRICULTURE

SMART AGRICULTURE.

SMARTER DECISIONS.

AI-Powered Insights. Farmer-Friendly Advice.



REAL-TIME MONITORING
Every field, every day

PRECISE INSIGHTS
Data-driven intelligence

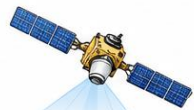
TIMELY ACTION
Right advice, right time

IMPROVED PRODUCTIVITY
Higher yield & quality

WATER EFFICIENCY
Save water, save future

TRUSTED TECHNOLOGY
Secure, scalable & reliable

AI FOR AGRICULTURE. INTELLIGENCE FOR IMPACT.™ Better Decisions. Better Farms. Better Future.



JALACHARA

AI-POWERED WATER INTELLIGENCE FOR EQUITABLE AGRICULTURE

Data → Intelligence → Fair Water → Sustainable Future



Smart Decisions.
Equal Water.
Stronger Farms.
Sustainable Future.



ISRO SATELLITE IMAGERY & DATA

- Optical Imagery (Resourcesat, EOS)
- SAR Data (Upcoming NISAR)
- Weather & Climate (IMD, GFS)
- Ground & IoT Sensors (Soil, Water, Flow)

1 CROP MAPPING

Accurate, timely and field-level crop identification

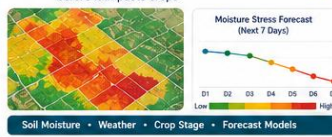


DATA PIPELINE



2 WATER DEFICIT FORECASTING

Predict future moisture stress before it impacts crops



8 EXPLAINABLE AI DASHBOARD

Transparent insights with clear explanations



JALACHARA AI ENGINE

Machine Learning | Deep Learning
Geospatial AI | Hydrological Models
Optimization | Fairness Algorithms

3 YIELD LOSS PREVENTION

Quantify risk and recommend timely interventions



7 POLICY SIMULATOR

Simulate policies and scenarios before implementation



ACTIONABLE INSIGHTS



4 WATER EQUITY INDEX

Measure fairness in water distribution across regions



6 WATER ALLOCATION OPTIMIZATION

Optimize canal releases and field-level allocation



5 EARLY WARNING SYSTEM

Alerts for drought, stress, floods and anomalies



OUTCOMES

- Higher Productivity
- Water Savings
- Cost Reduction
- Fair Water Access
- Climate Resilience
- Sustainable Future

STAKEHOLDERS

- Farmers
- Water Managers
- Government
- Researchers
- NGOs & Institutions
- Agri Businesses



DATA-DRIVEN



AI-POWERED



EQUITABLE



SCALABLE



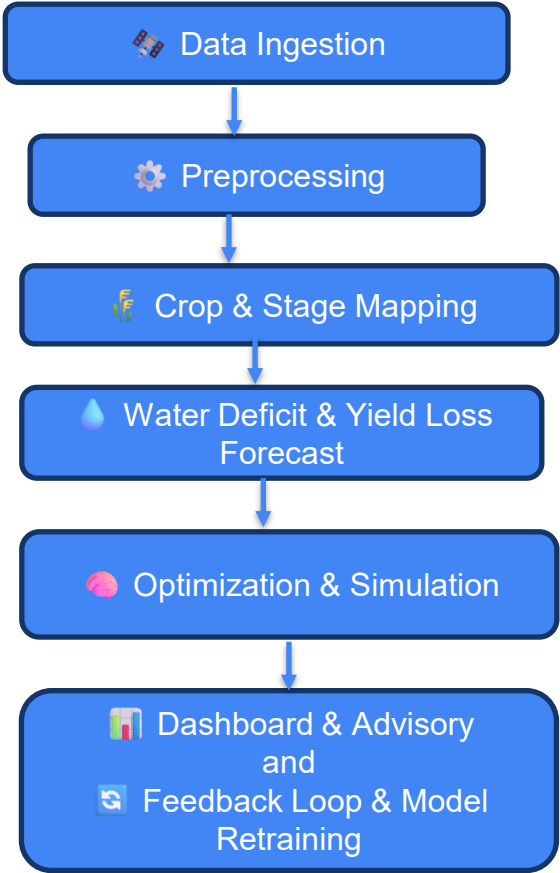
SUSTAINABLE



Jalachara – End-to-End Process Flow

Stage	Process	Output
 Data Ingestion	EOS-04, LISS-III/IV, Sentinel-1/2, ERA5-Land, AgriFieldNet	Raw satellite and weather data
 Preprocessing & Feature Extraction	Cloud masking, speckle filtering, NDVI, NDWI, EVI, SAR features	Clean feature dataset
 Crop & Growth Stage Mapping	SITS-Former classification	Crop type + Growth stage
 Water Deficit & Yield Loss Forecast	Physics-Informed ML + FAO-56	Water deficit (mm) + Yield loss (%)
 Optimization & Simulation	NSGA-II, Water Equity Index, Early Warning, What-If Simulator	Water allocation strategy
 Output & Advisory	Dashboard, Reports, SMS, Mobile Advisory	Actionable irrigation recommendations

Transforms multi-source satellite observations into predictive irrigation intelligence, equitable water allocation, and actionable farmer advisories.





JALACHARA

AGRICULTURAL AI PLATFORM



From Earth Observation to Intelligent Decisions



ENABLING TECHNOLOGIES



AI/ML & Deep Learning



Cloud Computing & Scalable Infra



Geospatial Analytics



Big Data Processing & Data Lake



APIs & Microservices



MLOps & Model Monitoring



Security & Data Governance

OUTCOMES & IMPACT



Optimized Water Use



Higher Crop Productivity



Equitable Water Distribution



Reduced Yield Loss



Resilient & Sustainable Agriculture



Data-driven Policy & Planning

Wireframes – How Jalachara Looks and Feels



JALACHARA
Satellite Monitoring for Equitable Agriculture

18 May 2025

Admin User
Dashboard

Dashboard
Map View
Fields
Crop Monitoring
Water Analytics
Alerts
Advisories
Reports
Policy Simulator
Settings
Help & Support

DATA SOURCES

- ISRO Satellite (EOS)
- Weather Data (IMD)
- Soil Moisture Sensors
- IoT Field Sensors
- Canal & Flow Data

Fields
Layers
Legend
Fullscreen



TOTAL FIELDS
1,248
+18 this week

AVG. WATER DEFICIT
23%
Moderate
-5% vs last week

WATER EQUITY INDEX
0.78 /100
Good
+0.08 vs last week

EARLY WARNING ALERTS
6
Active Alerts
View All

WATER DEFICIT OVERVIEW

WATER DEFICIT TREND

EARLY WARNING ALERTS

Alert Type	Location	Severity	Risk	Valid Until
Moisture Stress	Field Block 12	High	Crop at risk in 5-7 days	20 May 2025
Rainfall Deficit	Field Block 07	Moderate	Rainfall below normal	21 May 2025
Heat Stress	Field Block 15	Moderate	High temperature trend	22 May 2025
Irrigation Advisory	Field Block 03	Low	Irrigation recommended	19 May 2025

CROP HEALTH STATUS

TOP AT-RISK FIELDS

RECENT ADVISORIES

Last Data Update
18 May 2025, 08:30 AM

Next Satellite Overpass
19 May 2025, 10:45 AM

Weather Forecast
Light Rain, 28°C 19 May

System Status
All Systems Operational

Data-Driven Decisions. Equitable Water.
Sustainable Agriculture.™



JALACHARA

Satellite Intelligence for Equitable Agriculture

Search fields, locations...



18 May 2025



Admin User
Dashboard



Dashboard



Map View



Fields



Crop Monitoring



Water Analytics



Alerts



Advisories



Reports



Policy Simulator



Settings



Help & Support

DATA SOURCES



ISRO Satellite (EOS)



Weather Data (IMD)



Soil Moisture Sensors



IoT Field Sensors



Canal & Flow Data

← Back to Fields

Field Block 07

Village Rampur • District Nanded • Area 4.25 ha • Field ID FB-07



Crop

Rice

Variety: MTU 1010



Growth Stage

Flowering

(45-65 DAS)



Last Satellite Pass

18 May 2025

10:15 AM



Overall Field Health

Moderate

Score: 62 / 100

Field Overview

Advisories

Forecast

Explainable AI

History

Sensors

Documents



WATER DEFICIT

23%

Moderate

-5% vs last week



EXPECTED YIELD LOSS

14%

Without Action

High Risk



IRRIGATION RECOMMENDATION

Irrigate 35 mm

within 48 hours

Optimal Window

19 May - 20 May 2025

[View Details](#)



SOIL MOISTURE (0-30cm)

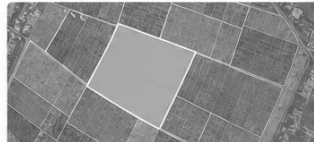
18%

Low

Ideal: 25-35%



FIELD MAP (NDVI)



Imagery Date: 18 May 2025

[View on Map](#)

- Very High
- High
- Moderate
- Low
- Very Low

8-DAY WATER DEFICIT FORECAST

DAY 1

18 May



22%

DAY 2

19 May



28%

DAY 3

20 May



34%

DAY 4

21 May



38%

DAY 5

22 May



41%

DAY 6

23 May



45%

DAY 7

24 May



47%

DAY 8

25 May



44%

Risk Level

Low Risk

Moderate Risk

High Risk

WHY THIS ADVICE?

AI analysis indicates increasing moisture stress over the next 3 days due to high evapotranspiration and low rainfall probability.

Key Factors



High Temperature

Forecast max 35-36°C



Low Rainfall Probability

10% chance in next 3 days



Soil Moisture Depletion

Current 18% (low)



Crop Stage Sensitivity

Flowering stage is highly sensitive

Moisture Stress Forecast vs Threshold



Confidence Score



High Confidence

Model Accuracy

85%

Data Quality

Good

CURRENT WEATHER



28°C

Light Rain

Humidity 72%

Wind 12 km/h

Rainfall (24h) 6.2 mm

SOIL SUMMARY



Texture

Loam

EC

0.48 dS/m

pH

6.8

Organic Carbon

0.62%

RECOMMENDATIONS



Irrigate 35 mm within 48 hours

Apply irrigation between 19 May - 20 May 2025



Maintain 2-3 cm standing water after irrigation

Ensure field leveling and proper bunding



Avoid nitrogen application before irrigation

Helps in better nutrient use efficiency



Monitor field regularly

Check for pests and disease during flowering stage

[View Full Advisory Report](#)



Last Data Update
18 May 2025, 10:30 AM



Next Satellite Overpass
19 May 2025, 10:45 AM



Weather Alert
Light Rain Expected 20 May



System Status
All Systems Operational



"Smart decisions today,
Sustainable harvests tomorrow."

Water Allocation What-If Simulator

Simulate and compare water allocation strategies across canal commands

1 Configure Scenario

Adjust parameters to create a scenario

Water Availability



Allocation Priorities

Set priority weights (Total must equal 100%)



Scenario Options

- ☒ Enable Deficit Irrigation
- ☒ Prioritize Drought Affected Areas
- ☒ Maintain Minimum Flow in Rivers
- ☐ Limit Maximum Allocation per Field

Advanced Settings

Reset

Run Simulation



Scenario Insight

Optimized scenario improves equity and yield while reducing average water deficit by 12% and saving 18 MCM of water.

2 Scenario Comparison

Create New Scenario

Baseline Scenario

Existing water release and distribution pattern

Total Water Available: 100 MCM

Current Strategy

Optimized Scenario

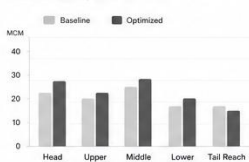
AI-optimized allocation based on your inputs

Total Water Available: 100 MCM

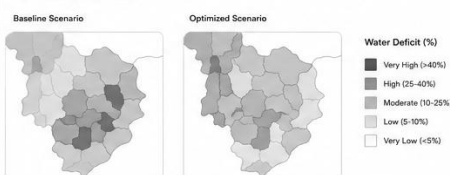
Key Outcome Comparison

		Baseline	Optimized	Change
Total Water Released	MCM	100 MCM	100 MCM	—
Average Water Deficit	%	28% (Moderate)	16% (Low)	▼ 12%
Total Yield	Tonnes	12,450 t	13,920 t	▲ 11.8%
Average Yield Increase	%	—	+11.8%	—
Water Savings	MCM	—	18 MCM	▲ 18%
Water Productivity	kg/ha	1.25	1.48	▲ 18.4%
Equity Index	(0-1)	0.62 (Moderate)	0.86 (High)	▲ 0.24

Water Allocation by Canal Zone (MCM)



Water Deficit Map Comparison



Recommendations

- ☒ Increase allocation to drought-affected tail reach by 8%
- ☒ Maintain environmental flow at minimum 10 MCM
- ☒ Monitor soil moisture and adjust mid-season if needed

Download Report

9:41

JALACHARA

Smart Water, Stronger Farms.

Village Rampur, District Nanded

IRRIGATION ALERT

Irrigate 35 mm within 48 hours

Optimal window: 19 May - 20 May

FIELD SUMMARY

Field ID: FB-07

Area: 4.25 ha

Crop: Rice (MTU 1010)

Growth Stage: Flowering (45-65 DAS)

CURRENT STATUS

Updated: 18 May 2025, 10:15 AM

Water Deficit: 23% Moderate

Soil Moisture: 18% Low

Field Health: 62/100 Moderate

YIELD LOSS WARNING

14% Expected Yield Loss Without Action

8-DAY WATER DEFICIT FORECAST

Today: 23%

Mon: 28%

Tue: 34%

Wed: 38%

Thu: 41%

Fri: 45%

Sat: 47%

Sun: 44%

View Full Advisory

Home Fields Advisories Alerts More

9:41

JALACHARA

Smart Water, Stronger Farms.

Village Rampur, District Nanded

Field ID: FB-07

Overview Irrigation Forecast AI Insights

IRRIGATION RECOMMENDATION

Irrigate 35 mm within 48 hours

Optimal window: 19 May - 20 May 2025

Next Irrigation: 19 May

Irrigation Amount: 35 mm

Duration: 6-8 hours

WHY THIS ADVICE?

AI analysis shows increasing moisture stress due to high temperature and low rainfall expected in the next 3 days.

View Details

KEY FACTORS

High Temperature: 35-36°C

Low Rainfall Probability: 10%

Soil Moisture Depletion: 18% (Low)

Crop Stage Sensitivity: Flowering stage

RECOMMENDATIONS

Irrigate 35 mm within 48 hours

Apply irrigation between 19 May - 20 May

Maintain 2-3 cm standing water after irrigation

Ensure field leveling and proper bunding

Avoid nitrogen application before irrigation

Helps in better nutrient use efficiency

Monitor field regularly

Check for pests and disease during flowering stage

VOICE SUPPORT

Tap to speak with Jalachara

Home Fields Advisories Alerts More

Technologies to be used in the solution:

Category	Technology / Tool	Purpose
 Data Ingestion	Google Earth Engine (GEE) Python API	Access Sentinel-1/2, MODIS, Landsat, ERA5-Land
 Data Ingestion	ISRO Bhoonidhi Portal	Download EOS-04, LISS-III, LISS-IV data
 Data Ingestion	Bhoonidhi Downloader	Automated ISRO data retrieval
 Preprocessing	eemont	Cloud masking, atmospheric correction
 Preprocessing	polsartools	SAR speckle filtering (Refined Lee)
 Preprocessing	SciPy (Savitzky-Golay Filter)	NDVI/EVI time-series smoothing
 Preprocessing	GDAL / Rasterio	Raster processing and reprojection
 AI/ML Models	PyTorch	SITS-Former crop and growth-stage classification
 AI/ML Models	XGBoost	Baseline crop classification
 AI/ML Models	Scikit-learn	Model training, validation, performance metrics
 AI/ML Models	Optuna	Hyperparameter optimization
 Optimization	pymoo (NSGA-II)	Multi-objective water allocation optimization

 Explainability	SHAP	Explainable AI and feature importance
 Physical Models	pyfao56	FAO-56 water balance and irrigation modeling
 Physical Models	AquaCrop-OSPy (Optional)	Crop water productivity validation
 Dashboard	Streamlit	Interactive web dashboard
 Dashboard	Plotly	Interactive charts and analytics
 Dashboard	Leafmap / Folium	Geospatial visualization and mapping
 Dashboard	ReportLab	Automated PDF advisory generation
 Deployment	Hugging Face Spaces	Dashboard hosting
 Deployment	GitHub	Source code management
 Deployment	GitHub Actions	CI/CD and automation
 Edge Delivery	TensorFlow Lite	Offline mobile deployment
 Edge Delivery	Flutter (Optional)	Mobile application development
 Edge Delivery	Twilio	SMS and WhatsApp advisory delivery
 Offline AI	Gemma	Native language offline advisory assistant

Technology Stack: 100% Open Source • ISRO Data Driven • Scalable • NISAR Ready • Cloud + Edge Deployment   

Architecture of Jalachara



DATA SOURCE LAYER

Multi-source data ingestion for comprehensive agricultural intelligence

SATELLITE & REMOTE SENSING DATA



WEATHER & CLIMATE DATA



GROUND TRUTH & AGRICULTURAL DATASETS

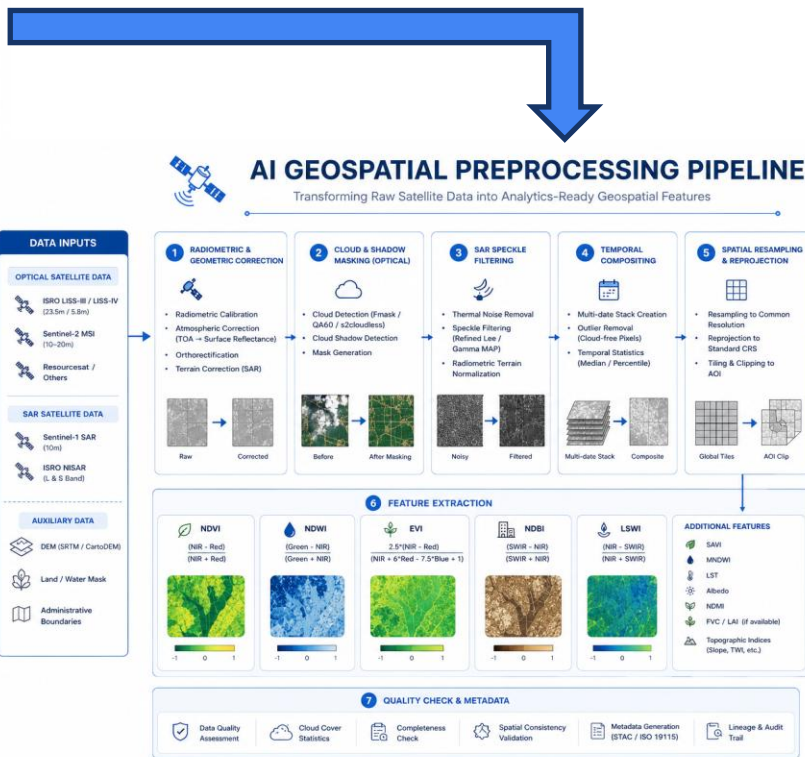


DATA INGESTION & PROCESSING PIPELINE

Quality Check • Standardization • Geospatial Alignment
Time Sync • Feature Engineering

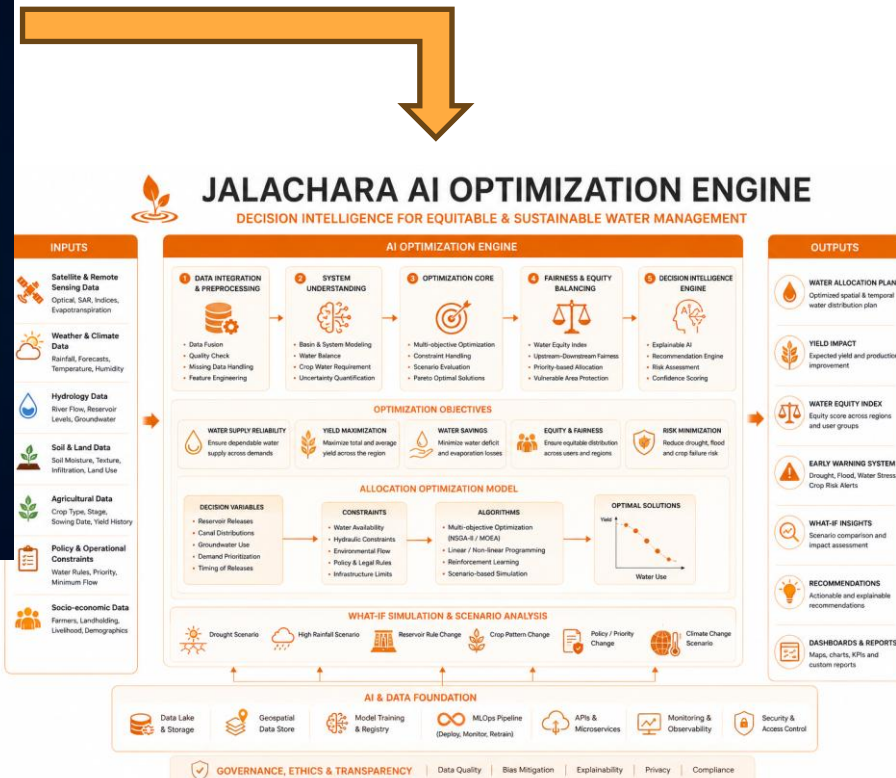
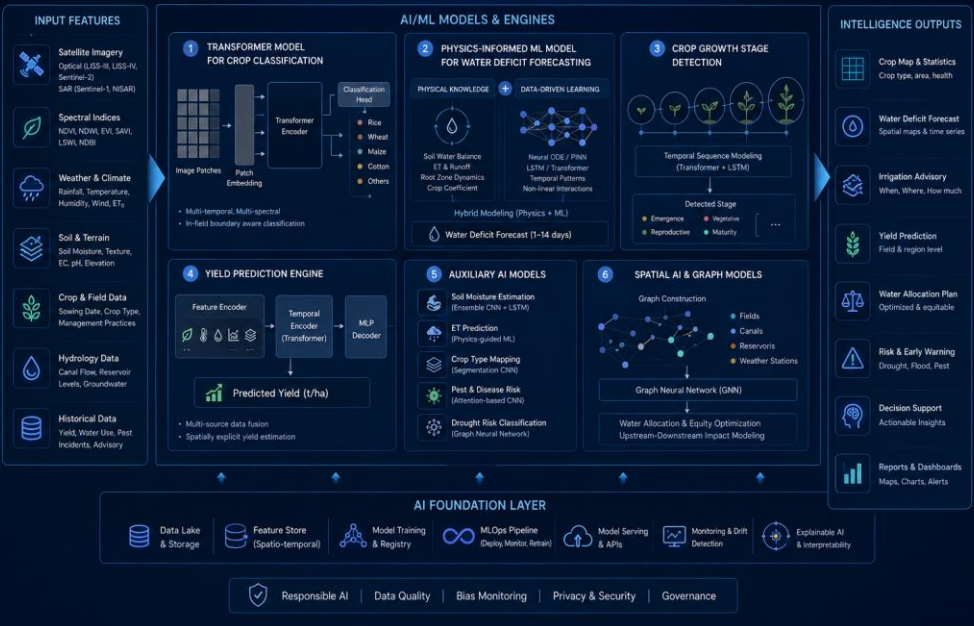
AI/ML MODELS & ANALYTICS ENGINE

Prediction • Optimization • Insights



JALACHARA AI ARCHITECTURE

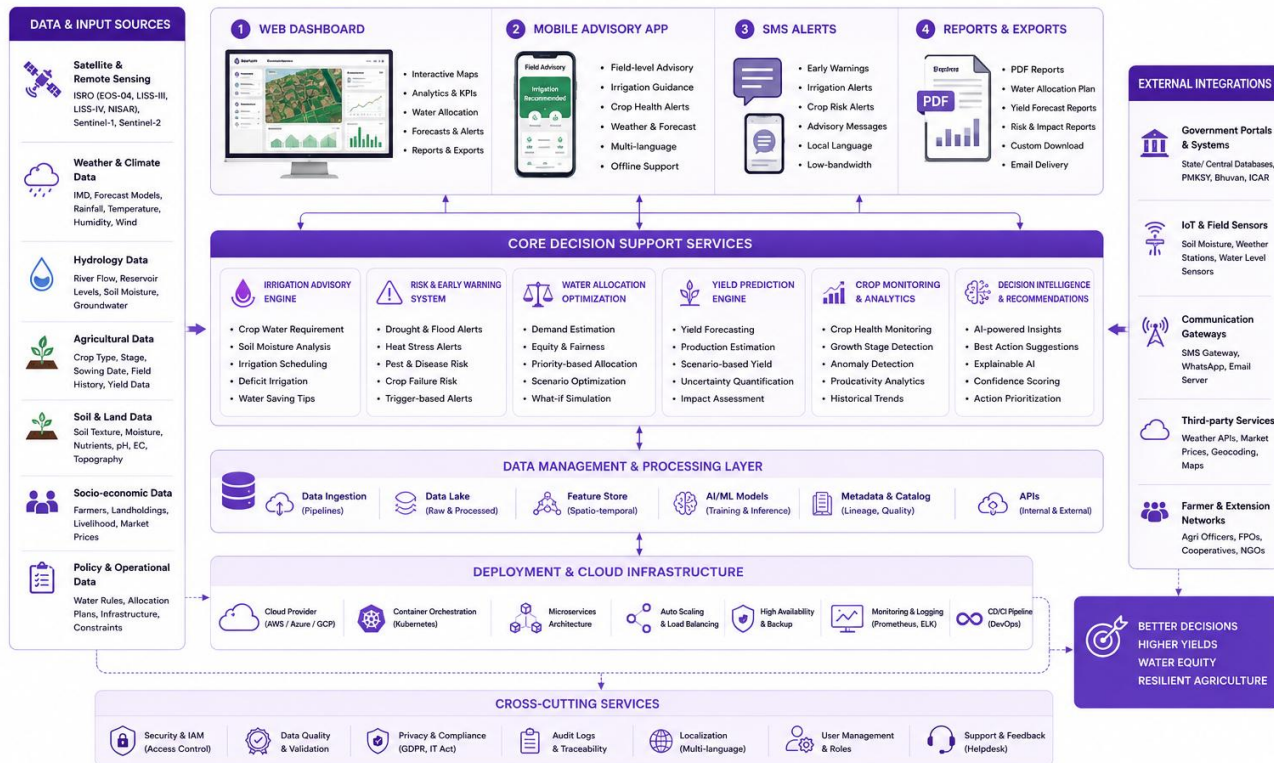
AI & ML ENGINE FOR INTELLIGENT AGRICULTURAL DECISIONS





JALACHARA AGRICULTURAL DECISION SUPPORT PLATFORM

From Data to Decisions for Equitable & Sustainable Agriculture



Modular,
scalable,
NISAR-ready
architecture
that transforms
satellite
observations
into predictive
irrigation
intelligence.

Estimated Implementation Cost

Component	Technology / Service	Cost
Satellite Data	ISRO Bhoonidhi (EOS-04, LISS-III/IV)	₹0
Satellite Data	Google Earth Engine (Sentinel-1/2, MODIS, ERA5-Land)	₹0
AI/ML Development	Python, PyTorch, XGBoost, SHAP, pymoo	₹0
Model Training	Google Colab (GPU)	₹0
Code Repository	GitHub	₹0
Dashboard Hosting	Hugging Face Spaces	₹0
SMS/WhatsApp Demo	Twilio Trial Credits	₹0
Mobile Deployment	Flutter + TensorFlow Lite	₹0
Development Team	Student Team	₹0

Estimated Cost for Real-World Deployment

Component	Annual Cost (Approx.)
Cloud Compute Infrastructure	₹2–5 Lakhs
Data Storage & Backup	₹1–2 Lakhs
Dashboard & API Hosting	₹1–3 Lakhs
SMS/WhatsApp Advisory Services	₹1–2 Lakhs
Maintenance & Model Updates	₹5–10 Lakhs

Jalachara can be developed at zero cost during the hackathon using free ISRO datasets, open-source AI frameworks, and free cloud resources. Even at national scale, the operational cost is minimal compared to the potential benefits of water conservation, yield-loss prevention, and equitable irrigation management.

BHARATIYA
ANTARIKSH
HACKATHON

2026

THANK YOU